Probe-Gated Critic Transfer

Latent Team Discovery from Individual Reward Responses

[Presenter] · [Affiliation]

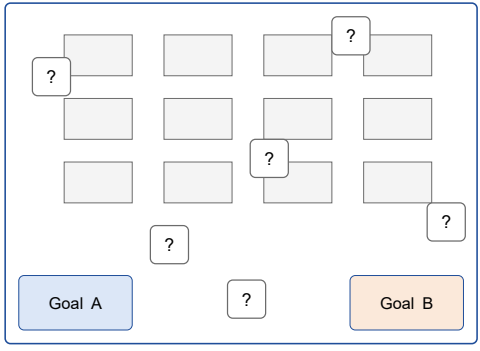
Research presentation draft · November 2026

Outline

- Introduction
- Proposed Method
- Simulation Results
- Conclusion

Introduction: Research Background

- Shared workspace: several robot teams work in one warehouse
- Different goals: each team serves its own requests and goal
- Hidden teams: an agent sees its neighbors, not their team
- Core message: being close does not mean sharing a goal

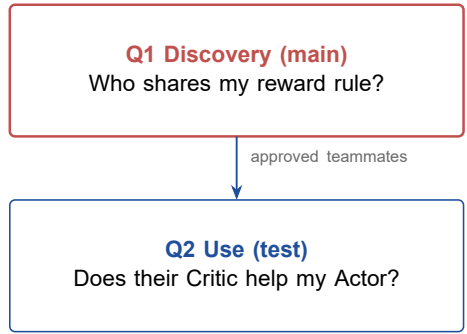


Who is my teammate?

robots look alike; team labels are hidden

Introduction: Problem Definition

- Given: own observation, own action, own reward
- Not given: team labels and other agents' rewards
- Communication: only on contact (within 3 cells)
- Q1 Discovery: find latent teammates online
- Q2 Use: does teammate value information help learning?



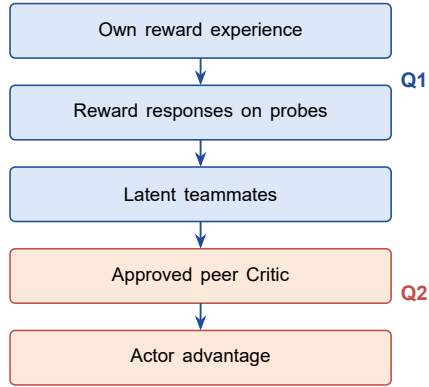
Introduction: Related Work

Approach	Strength	Limitation
Independent PPO [Tan 1993; de Witt 2020]	simple, strong baseline	no help from teammates
Shared experience / parameters [SEAC 2020; SePS 2021]	faster learning in a group	the group must be known
Consensus Critic [Zhang 2018]	fully decentralized	one common objective
Max over several values [GPI 2017; MAMBA 2020]	combines better values	values must be valid in my states

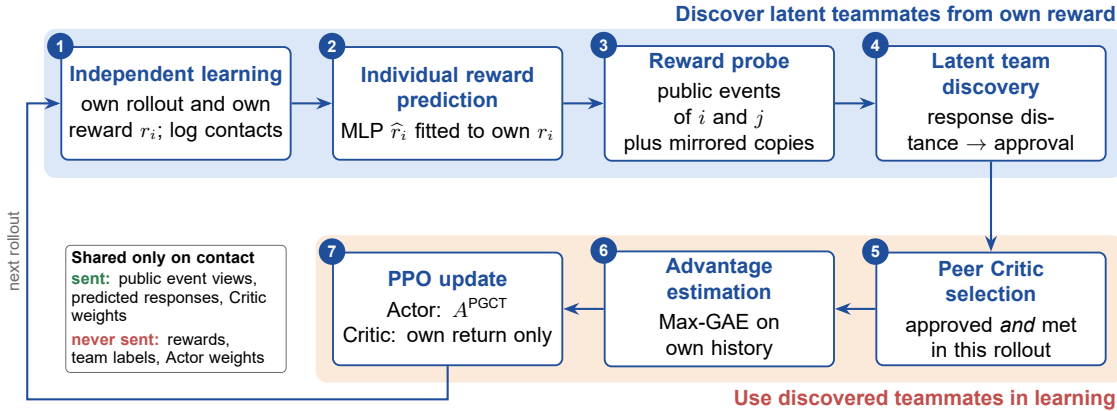
Research gap: no method finds teammates online from individual rewards and then uses them selectively.

Introduction: Research Goal and Contributions

- Objective: discover latent teammates from individual reward experience
- Key idea: same team $\rightarrow$ same reward rule $\rightarrow$ same reward responses
- Contribution 1: Reward Probe test with abstention
- Contribution 2: causal, contact-limited discovery
- Contribution 3: selective peer Critic use for the Actor only



Proposed Method: Overall PGCT Architecture

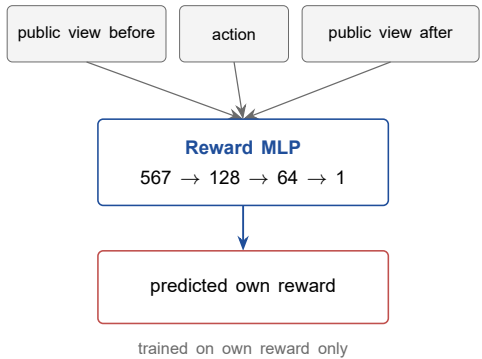


All steps are local. Peers exchange information only on contact.

Proposed Method: Individual Reward Prediction

- Input: public view before, action, public view after
- Public view: no requested shelves, no own goal, no coordinates
- Target: own reward only
- Sparse events: deliveries and returns ($r \geq 0.5$): 25% of each batch

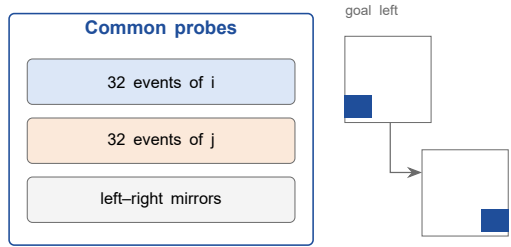
$$\widehat{r}_i(q) = \text{MLP}_{\psi_i}\left[u_t,\ \text{onehot}(a_t),\ u_{t+1}\right], \quad u_t = P(o_{i,t})$$



Proposed Method: Reward Probe Construction

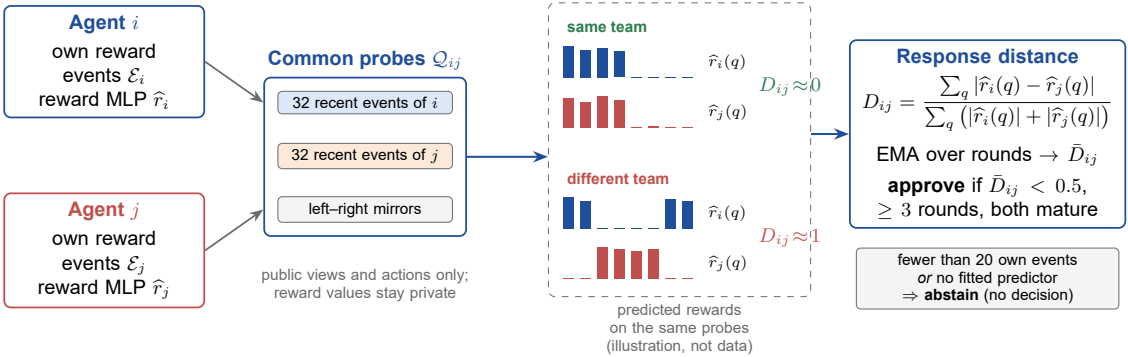
- Probe: one public reward event: before, action, after
- From both agents: 32 recent own events each
- Mirror: flip left–right: asks about the other goal
- Snapshot: probes are fixed at the rollout start
- Never shared: reward values and team labels

$$\mathcal{Q}_{ij}^{u} = \mathcal{E}_{i}^{u} \cup \mathcal{E}_{j}^{u} \cup M\left(\mathcal{E}_{i}^{u} \cup \mathcal{E}_{j}^{u}\right)$$



The same probes go to both predictors.
Only public views and actions are shared.

Proposed Method: Reward Response Distance and Team Discovery

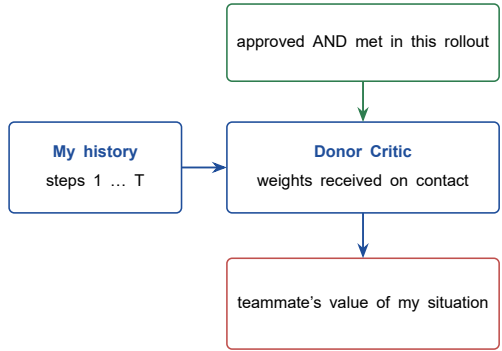


$$B_{ij} = \mathbf{1}\left[\, n_{ij} \geq 3,\ \bar{D}_{ij} < 0.5,\ m_i = m_j = 1 \,\right]$$

Not mature (fewer than 20 own events) $\rightarrow$ abstain. No team label is used.

Proposed Method: Selective Peer Critic Access

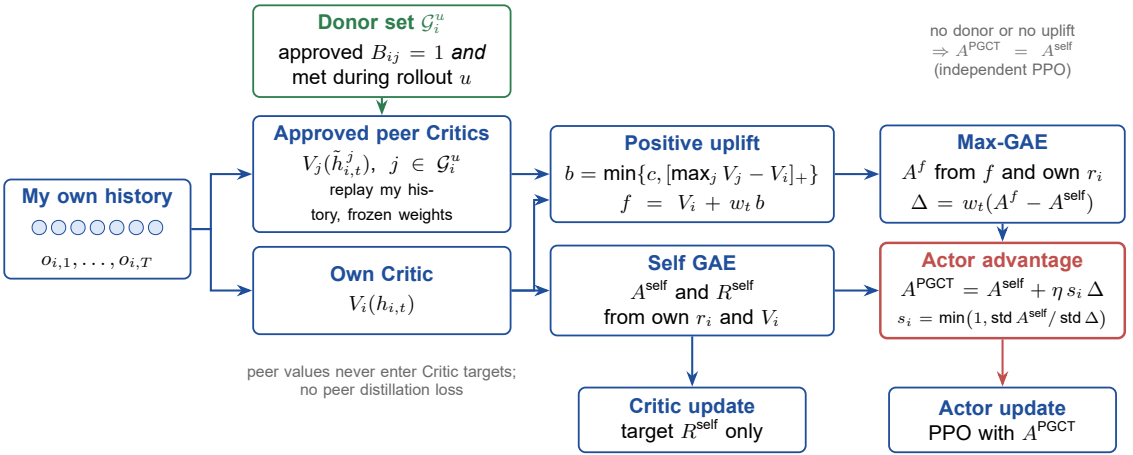
- Donor: approved and met in this rollout
- Receive: Critic weights on contact (frozen)
- Replay: my own history through the donor Critic
- Not used: donor hidden state, data or rewards
- No donor: exactly independent PPO



donor hidden state is never sent

$$\mathcal{G}_i^u = \{\, j \neq i : \; C_{ij}^u = 1, \; B_{ij} = 1 \,\}$$

Proposed Method: Max-GAE Advantage Estimation

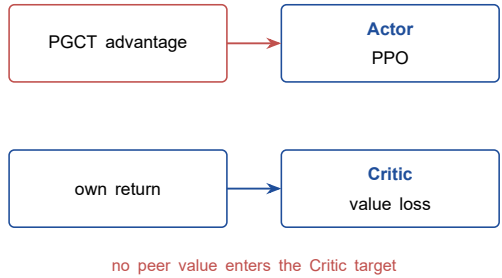


$$A_{i,t}^{\text{PGCT}} = A_{i,t}^{\text{self}} + \eta\, s_i\, w_t \big(A_{i,t}^{f} - A_{i,t}^{\text{self}}\big), \quad s_i \le 1$$

Cold-start steps are masked (w = 0). The scale s $\le$ 1 only shrinks; it never inflates.

Proposed Method: Actor and Critic Updates

- Actor: PPO with the normalized PGCT advantage
- Critic: own return only; no distillation loss
- Separate networks: own reward is never changed
- Safe default: no donor or no uplift → independent PPO



$$L_i^{\pi} = \mathbb{E}_t\left[\min\{\rho_{i,t}\widetilde{A}_{i,t},\ \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)\widetilde{A}_{i,t}\}\right]$$

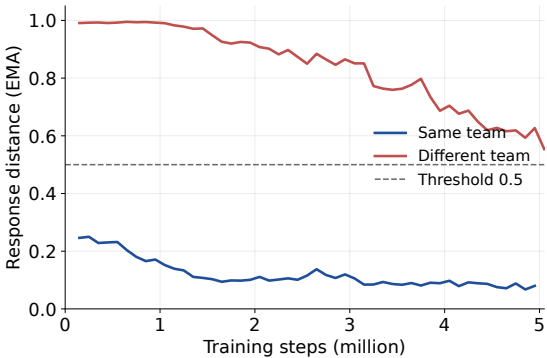
$$R_{i,t}^{\text{self}} = A_{i,t}^{\text{self}} + V_i^{\text{old}}(h_{i,t})$$

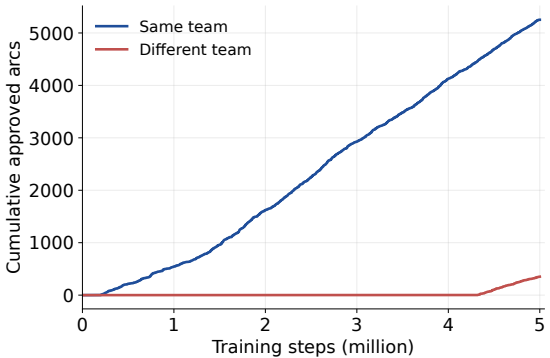
Simulation Results: Experimental Setup

- Environment: multi-team robotic warehouse, 11×10, 6 agents (3 + 3)
- Goals: one per team, at the two bottom corners
- Reward: individual: delivery 1, return 1, pickup 0.15, small shaping
- Methods: Independent PPO · True-team Max-GAE · True-team (mature only) · PGCT · Count-matched random
- Protocol: branch from a common 1.024M checkpoint, +0.5M steps, seeds 1, 3, 5

Setting	Value
Grid / agents	11×10 / 6 (3 + 3)
Sensor / contact radius	5 / 3 cells
Rollout / γ / λ	1024 / 0.99 / 0.95
Event / maturity	r $\geq$ 0.5 / 20 events
EMA β / threshold / rounds	0.35 / 0.5 / 3
Max-GAE η / scale	0.5 / shrink only
Evaluation	20 episodes $\times$ 500 steps

Simulation Results: Team Discovery Performance

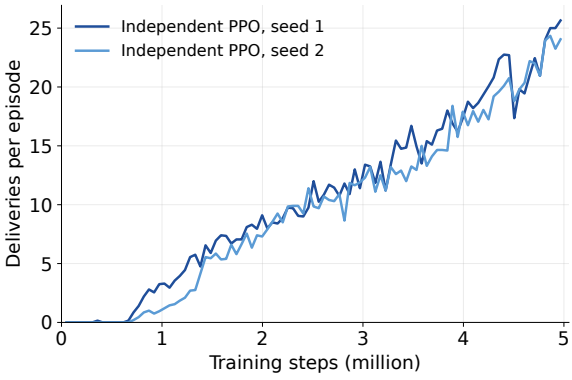




- Early separation: 0.1–0.25M: distance 0.25 (same) vs 0.99 (different)
- Correct access: 93.7% to true teammates (5,252 vs 352 arcs)
- First cross-team approval: only at 4.28M steps

Corner goals, individual rewards, seed 1, 5M steps. Discovery component only (update-end contacts).

Simulation Results: Learning Performance



Counterfactual check (why teams matter)

- Same-team uplift > 0.2: real gain in 67–100% of cases
- Different-team uplift: mean gain only −0.01 to 0.14

Method (+0.5M)	Lagging	Leading
Independent PPO	*Pending*	*Pending*
True-team Max-GAE	*Pending*	*Pending*
True-team, mature only	*Pending*	*Pending*
PGCT	*Pending*	*Pending*
Count-matched random	*Pending*	*Pending*

Branch runs in progress. Pending cells will be replaced by final results; no partial curves are shown.

Conclusion

- Summary: PGCT finds latent teammates from individual reward responses
- Main contribution: Reward Probe test: local, causal, abstains without evidence
- Use: approved peer Critics shape the Actor only; Critics keep own return
- Evidence: early team separation; 93.7% correct peer access
- Limitation: agents without reward events cannot be matched
- Future work: finish the learning comparison; more seeds and environments

Thank you! Questions?